

25% Sample vs. Full-Fleet Equivalence

UrbanEV–Maryland paired-run comparison (preliminary, full run in progress)

UrbanEV–Maryland verification series

July 24, 2026

Purpose

All reported results run on a deterministic hash-based 25% sample of the 148,302-owner EV fleet, with charger plug counts scaled by 0.25 (floor one), flow capacity 0.25, storage capacity $0.25^{0.75} \approx 0.35$, and fleet quantities scaled back by four. This note checks that design against a paired full-fleet run of the identical model: same plans (pre-sampling), same charger network at full plug counts, same prices, same AADT-loaded network, same 50-iteration protocol. The full run is still executing; the comparison below uses its latest completed iteration (34 of 50) against the converged 25% baseline, and the definitive table will be produced from the full run’s converged output.

Preliminary comparison

Metric	25% $\times$ 4 (converged)	Full fleet (in progress)	Difference
Charging agents (fleet)	69,652	78,523	+12.7%
Sessions per day (fleet)	45,376	47,392	+4.4%
Home session share	78.1%	77.4%	−0.7 pp
Work session share	13.4%	11.4%	−2.0 pp
Public L2 session share	8.2%	9.8%	+1.6 pp
DCFC + Tesla session share	0.34%	0.41%	+0.07 pp
Home energy share	81.6%	82.2%	+0.6 pp
Public energy share	5.7%	6.9%	+1.2 pp
Annual charged energy	374 GWh	389 GWh	+4.3%
ChargePoint occupancy r	0.773	0.842	+9.0%
Session-start r (24 h, weekday)	0.934	0.936	+0.2%
Session-start r (daytime-only)	0.833	0.848	+1.8%

Reading

The structure the results depend on transfers. Home dominance matches within one percentage point of energy share, total session volume and annual energy agree within about $4\frac{1}{2}\%$, and both timing validations hold or improve at full scale: the occupancy shape rises from $r = 0.77$ to 0.84, while the session-start profile is essentially scale-invariant ($r = 0.93$ – 0.94 over 24 hours; 0.83 – 0.85 daytime-only) — with four times the agents, the simulated diurnal profiles are smoother and track the ChargePoint panel at least as closely. The sampling design therefore does not manufacture the paper’s findings; if anything the 25% run is the noisier of the pair.

The residual gaps sit where mid-run drift lives. The public-L2 (+1.6 pp) and work (−2.0 pp) session-share gaps match the L2–work drift documented in the 25% baseline at the same stage of convergence, which settled over the final fifteen iterations there; the same narrowing is expected here as innovation switches off and selection completes. The larger count of unique charging agents (+12.7%) typically declines with convergence for the same reason.

No conclusion is at risk under either endpoint. Taking the in-progress values at face value, the public energy base would be 6.9% instead of 5.7%; the public-surcharge revenue ceiling would scale proportionally to roughly \$11M, still about one third of $R^* = \$33.3\text{M}$, leaving the structural-inadequacy conclusion untouched. The home surcharge, distance-based, and incidence results are insensitive at these magnitudes.

Use in the manuscript

The manuscript states that paired runs verified sample-to-fleet equivalence on charger-type shares. The quantitative sentence will cite the full run’s converged output (no interim iteration values), expected within days and before submission: home energy share within one percentage point, fleet totals within $\sim 5\%$, occupancy correlation preserved or improved, venue shares within ~ 1.5 percentage points, and the full fleet’s slightly larger public share only strengthening the conservatism of the public-surcharge ceiling.